OSTIn project: Develop software to automatically launch infrasound and satellite imagery analyses using the time and location provided by a VONA (Volcano Observatory Notice for Aviation).

Parse VONA messages to extract time and location data from both MAGMA (MAGMA VONA) and TVAAC (TVAAC VONA) in real time

Sources: MAGMA VONA: https://magma.esdm.go.id/v1/vona

TVAAC VONA: https://www.data.jma.go.jp/svd/vaac/data/vaac_list.html

Forecast Eruption Source Paramerters such as Mass Eruption Rate, Total Erupted Mass, Eruption Duration and percentage of fine ash particles based on VONA messages and Historical Volcanic Eruption Data using Bayesian statistics

VONA Monitoring Script

This script continuously monitors the latest Volcanic Observatory Notice for Aviation (VONA) reports and detects changes in key volcanic activity parameters such as volcano name, ash cloud height above ground level (AGL), and volcanic explosivity index (VEI) range. When changes are detected, the screen is cleared, and updated information is displayed.

Key Features:

- Fetches VONA reports from a specified data source (default: "TOKYO").
- Detects changes in volcano name, ash cloud height, and VEI range.
- Updates the display when changes occur.
- Uses the `ESP_INIT` function for eruption source parameter analysis when new data is available.
- Implements a progress bar to visualize the monitoring interval.

Dependencies:

- `vona_trigger` : Retrieves VONA reports.
- `re` : For text normalization.
- `esp_init.ESP_INIT` : Processes eruption source parameters.
- `time` : Introduces delays between monitoring cycles.
- `os` : Clears the terminal screen.
- `tqdm` : Provides a visual progress bar for monitoring intervals.

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In [ ]:
```
"""
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"""


import vona_trigger
import re
from esp_init import ESP_INIT
import time
import os
from tqdm import tqdm  # For progress bar

# Initialize storage for previous variable values
previous_values = {
    "volcano_name": None,
    "AshCloudHeight_AGL": None,
    "vei_range": None
}
```

```python
def clear_screen():
    """Clears the terminal screen for a fresh output."""
    os.system('cls' if os.name == 'nt' else 'clear')

def normalize_text(text):
    """Removes extra spaces and normalizes text for meaningful comparison."""
    if isinstance(text, str):
        return re.sub(r'\s+', ' ', text).strip()  # Replace multiple spaces with one and trim
    return text  # Return non-string values as-is

while True:
    # Fetch the latest VONA data
    VONA_reports, VONA_latest_data, volcano_name, latitude, longitude, AshCloudHeight_AGL, vei_range = vona_trigger.VONA_Messa



    # Store current values in a dictionary
    current_values = {

        "volcano_name": volcano_name,
        "AshCloudHeight_AGL": AshCloudHeight_AGL,
        "vei_range": vei_range
    }

   # Check for changes in each variable
    changes_detected = False
    for key in current_values:
        prev_value = normalize_text(previous_values[key]) if previous_values[key] is not None else None
        curr_value = normalize_text(current_values[key])

        if prev_value != curr_value:
            changes_detected = True

# If changes are detected, clear screen and print new data

if changes_detected:
    clear_screen()  # Clears previous prints and outputs
    print("🔔 **New VONA Data Detected:**\n")
    for key in current_values:
        prev_value = normalize_text(previous_values[key]) if previous_values[key] is not None else None
        curr_value = normalize_text(current_values[key])

        if prev_value != curr_value:
            print(f"🔹 **Change detected in {key}:**")
            print(f"   - **Previous:** {prev_value}")
            print(f"   - **New:** {curr_value}\n")
            previous_values[key] = current_values[key]  # Update stored value

    # Convert AshCloudHeight_AGL to km if it's not None
    if AshCloudHeight_AGL is not None or vei_range is  None:

        ESP_INIT(vei_levels=None, PlumeHeight=AshCloudHeight_AGL, show_plots=True)

    if vei_range is not None or AshCloudHeight_AGL is  None:
        ESP_INIT(vei_levels=vei_range, PlumeHeight=None, show_plots=True)

    # if AshCloudHeight_AGL is not None and vei_range is not None:

    #     ESP_INIT(vei_levels=None, PlumeHeight=AshCloudHeight_AGL, show_plots=True)
    #     ESP_INIT(vei_levels=vei_range, PlumeHeight=None, show_plots=True)

    elif AshCloudHeight_AGL is  None and vei_range is  None:
        ESP_INIT(vei_levels=None, PlumeHeight=None, show_plots=True)

# Progress bar with dynamic status
status_message = "🟢 Waiting for next VONA report..." if changes_detected else "🟡 No new data - Monitoring..."

with tqdm(total=1800, desc=status_message, unit="s", bar_format="{l_bar}{bar} {remaining}") as pbar:
    for _ in range(1800):
        time.sleep(1)  # Sleep in 1-second intervals
        pbar.update(1)  # Update progress bar
```

```
CSV file created: ./TVAAC_test\vaa_texts.csv
Scraping up to 1 pages for All Volcanoes...
🔄 Scraping Pages: 100%|██████████| 1/1 [00:07<00:00,  7.16s/page]
Scraping completed. Data saved at Magma_Test\vona_reports_sorted.csv
Extracting details from 15 reports...
📝 Extracting Reports:   0%|          | 0/15 [00:00<?, ?report/s]2025-02-22 23:37:11,690 - WARNING - Connection pool is full, discarding connection: magma.esdm.go.id. Connection pool size: 10
2025-02-22 23:37:11,690 - WARNING - Connection pool is full, discarding connection: magma.esdm.go.id. Connection pool size: 10
2025-02-22 23:37:11,710 - WARNING - Connection pool is full, discarding connection: magma.esdm.go.id. Connection pool size: 10
2025-02-22 23:37:11,710 - WARNING - Connection pool is full, discarding connection: magma.esdm.go.id. Connection pool size: 10
2025-02-22 23:37:11,710 - WARNING - Connection pool is full, discarding connection: magma.esdm.go.id. Connection pool size: 10
📝 Extracting Reports: 100%|██████████| 15/15 [00:01<00:00, 12.98report/s]
💾 Saving extracted reports...
📁 Saving Reports: 100%|██████████| 15/15 [00:00<00:00, 188.03file/s]
```

✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0627Z_Ibu_(268030).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0538Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0536Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0438Z_Ibu_(268030).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0349Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0233Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0100Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250223-0027Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-2252Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-2131Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-2108Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-1926Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-1448Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-1413Z_Semeru_(263300).txt
✅ Data saved: Magma_Test\detailed_reports\VONA_Report_20250222-1345Z_Semeru_(263300).txt
Detailed reports extracted successfully!
Top VEI values returned: [2]

VONA Message Attributes from TVAAC VONA Center

Parameter	Value
Site of Volcanic Activity	KANLAON
Latitude	10.416666666666666
Longitude	123.13333333333334
Possible VEI Levels for Volcano KANLAON (Whelley et al. 2015)	[2]

Top VEI values returned: [2, 3, 4, 5, 6]

VONA Message Attributes from MAGMA VONA Center

Parameter	Value
Source URL	https://magma.esdm.go.id/v1/vona/15151?signature=8efeac09de335b0034…
Issued Date	20250223/0627Z
Volcano	Ibu (268030)
Current Aviation Colour Code	ORANGE
Previous Aviation Colour Code	orange
Observing Source	Ibu Volcano Observatory
Notice Number	2025IBU394
Volcano Location	N 01 deg 29 min 17 sec E 127 deg 37 min 48 sec
Area	North Maluku, Indonesia
Summit Elevation	4240 FT (1325 M)
Volcanic Activity Summary	Eruption with volcanic ash cloud at 0627 UTC (1527 local).
Volcanic Cloud Height	Best estimate of ash-cloud top is around 5840 FT (1825 M) above sea level or 1600 FT (500 M) above summit. May be higher than what can be observed clearly. Source of height data: ground observer.
Other Volcanic Cloud Information	Ash cloud moving to southwest. Volcanic ash is observed to be white to gray. The intensity of volcanic ash is observed to be thick.
Remarks	Eruption recorded on seismogram with maximum amplitude 28 mm and maximum duration 42 second.
Contacts	Center for Volcanology and Geological Hazard Mitigation (CVGHM).Tel: +62-22-727-2606.Facsimile: +62-22-720-2761.email : pvmbg@esdm.go.id
Next Notice	A new VONA will be issued if conditions change significantly or the colour code is changes.Latest Volcanic information is posted atVONA | MAGMA IndonesiaWebsite.Link:https://magma.esdm.go.id/v1/vona
Latitude	1.4880555555555557
Longitude	127.63
Ash Cloud Height (ASL)	1825 meters
Ash Cloud Height (AGL)	500 meters
Possible VEI Levels for Volcano Ibu (Whelley et al. 2015)	[2, 3, 4, 5, 6]

🔔 **New VONA Data Detected:**

🔹 **Change detected in volcano_name:**
- **Previous:** None
- **New:** Ibu

🔹 **Change detected in AshCloudHeight_AGL:**
- **Previous:** None
- **New:** 0.5

🔹 **Change detected in vei_range:**
- **Previous:** None
- **New:** [2, 3, 4, 5, 6]

```
Model updated with xvar='Height_km.a.v.l'
Model updated with yvar='MER_kg/s'.
Saved plot to ESP_for0.5km/Forecast_MER_kg_per_second\posterior_predictive_plot.png
Saved plot to ESP_for0.5km/Forecast_MER_kg_per_second\percentiles_plot.png
Saved plot to ESP_for0.5km/Forecast_MER_kg_per_second\contour_percentiles_plot.png
Saved DataFrame to ESP_for0.5km/Forecast_MER_kg_per_second\forecast_MER_kg_per_second.csv
Model updated with xvar='MER_kg/s'
Model updated with yvar='MER_kg/s'.
Model updated with yvar='TEM_kg'.
Saved plot to ESP_for0.5km/Forecast_TEM_kg\posterior_predictive_plot.png
Saved plot to ESP_for0.5km/Forecast_TEM_kg\percentiles_plot.png
Saved plot to ESP_for0.5km/Forecast_TEM_kg\contour_percentiles_plot.png
Saved DataFrame to ESP_for0.5km/Forecast_TEM_kg\forecast_TEM_kg.csv
Model updated with xvar='MER_kg/s'
Model updated with yvar='MER_kg/s'.
Model updated with yvar='Duration_hours'.
Saved plot to ESP_for0.5km/Forecast_Duration_hours\posterior_predictive_plot.png
Saved plot to ESP_for0.5km/Forecast_Duration_hours\percentiles_plot.png
Saved plot to ESP_for0.5km/Forecast_Duration_hours\contour_percentiles_plot.png
Saved DataFrame to ESP_for0.5km/Forecast_Duration_hours\forecast_Duration_hours.csv
Model updated with xvar='MER_kg/s'
Model updated with yvar='MER_kg/s'.
Model updated with yvar='MASS_FRACTION_ASH_BELOW_63_micron'.
Saved plot to ESP_for0.5km/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\posterior_predictive_plot.png
Saved plot to ESP_for0.5km/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\percentiles_plot.png
Saved plot to ESP_for0.5km/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\contour_percentiles_plot.png
Saved DataFrame to ESP_for0.5km/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\forecast_MASS_FRACTION_ASH_BELOW_63_micron.csv
Results saved in folder: ESP_for0.5km
Generating probabilistic volumes using Bayesian updating...
✅ Probabilistic volumes successfully generated within valid VEI bounds!
Calculating mass...
✅ Mass calculation completed!
Plotting volume percentile contours...
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]\plot_percentile_contours_VEIvolume.png
Plotting mass percentile contours...
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]\plot_percentile_contours_VEImass.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]\plot_posterior_samples_histograms_VEIvolume.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]\plot_posterior_samples_histograms_VEImass.png
✅ VEI model updated in place with filtered VEI levels, ensuring volumes remain within valid ranges.
Model updated with xvar='TEM_kg'
Model updated with yvar='MER_kg/s'.
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MER_kg_per_second\posterior_predictive_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MER_kg_per_second\percentiles_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MER_kg_per_second\contour_percentiles_plot.png
Saved DataFrame to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MER_kg_per_second\forecast_MER_kg_per_second.csv
Model updated with xvar='MER_kg/s'
Model updated with yvar='MER_kg/s'.
Model updated with yvar='Height_km.a.v.l'.
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_Height_km.a.v.l\posterior_predictive_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_Height_km.a.v.l\percentiles_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_Height_km.a.v.l\contour_percentiles_plot.png
Saved DataFrame to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_Height_km.a.v.l\forecast_Height_km.a.v.l.csv
```

VEI and Related Data

VEI	MER_kg/s	Height_km.a.v.l_Percentile_5	Height_km.a.v.l_Percentile_10
2.0	381035.5418068947	5.746341183694633	5.910252923033561
3.0	1997469.2162961396	8.911308310294011	9.091726990425228
4.0	10436897.98717856	13.430671696399841	13.692388695007942
5.0	54765330.913735405	19.62579805975959	20.09659505291818
6.0	287085981.3519558	28.157103478780304	29.122344065152884

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_15	Height_km.a.v.l_Percentile_20	Height_km.a.v.l_Percentile_25
2.0	6.027227023731285	6.116924159461623	6.1936946247564535
3.0	9.223869933822156	9.323725206369645	9.405899710206441
4.0	13.857854788800463	13.986437879314446	14.105161731311068
5.0	20.436508948924892	20.692514846423286	20.929121160111254
6.0	29.80967761826882	30.35577673217353	30.825319138386803

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_30	Height_km.a.v.l_Percentile_35	Height_km.a.v.l_Percentile_40
2.0	6.2648780675103835	6.331821170364636	6.3967812588565485
3.0	9.492200375894525	9.560597141712615	9.6328797841623
4.0	14.227394551100137	14.334047174825773	14.440595038406121
5.0	21.14610146093919	21.34785940038889	21.535854040278455
6.0	31.22317637818926	31.649241760087016	32.06745085116941

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_45	Height_km.a.v.l_Percentile_50	Height_km.a.v.l_Percentile_55
2.0	6.4566939581825995	6.521579649342344	6.5883424639188854
3.0	9.703704242714277	9.768726732753434	9.840281431061996
4.0	14.53529895536355	14.633471518133945	14.722660648871901
5.0	21.717876722551033	21.90284685097332	22.102215832838823
6.0	32.45791834423303	32.8604126435376	33.22874906403907

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_60	Height_km.a.v.l_Percentile_65	Height_km.a.v.l_Percentile_70
2.0	6.656123537917029	6.723017698976142	6.796560792131004
3.0	9.908590233207809	9.987237745070209	10.064925531084555
4.0	14.821879009399826	14.921384412309775	15.02736802040386
5.0	22.30627274345046	22.491961982282277	22.710610194321312
6.0	33.6061058388961	33.992640167299456	34.424051343581155

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_75	Height_km.a.v.l_Percentile_80	Height_km.a.v.l_Percentile_85
2.0	6.874141929611854	6.965880105310794	7.065561159854779
3.0	10.152725915116372	10.243828489473293	10.357316786138965
4.0	15.14884885961262	15.281360023745158	15.431934334286638
5.0	22.917408055544964	23.178419093285395	23.495993251069876
6.0	34.90189733348799	35.4459784302142	36.14323133145394

VEI and Related Data

VEI	Height_km.a.v.l_Percentile_90	Height_km.a.v.l_Percentile_95	Duration_hr_P5
2.0	7.205284446036452	7.429978648849934	3.649802925422445
3.0	10.499673219621524	10.743180623308469	6.987969129653975
4.0	15.638311284221862	15.965432111185953	15.708503955739335
5.0	23.883100566101817	24.527441821106187	37.19063417045742
6.0	36.99680887286935	38.29748080166311	89.31100337340503

VEI and Related Data

VEI	Duration_hr_P25	Duration_hr_P50	Duration_hr_P75
2.0	2.985009127624395	2.5847293495075667	2.2509456502513117
3.0	5.722233134843691	4.94212756896897	4.2711691335772
4.0	11.656139233838717	9.428279815429782	7.654472650036165
5.0	24.287981039081117	18.004557786621938	13.392289835144707
6.0	51.234426477815106	34.333445549853955	23.23904608116425

VEI and Related Data

VEI	Duration_hr_P95
2.0	1.8338988837298864
3.0	3.4829400618917434
4.0	5.656976129713682
5.0	8.6715265923139
6.0	12.985688970741975

```
Model updated with xvar='MER_kg/s'
Model updated with yvar='MER_kg/s'.
Model updated with yvar='MASS_FRACTION_ASH_BELOW_63_micron'.
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\posterior_predictive_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\percentiles_plot.png
Saved plot to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\contour_percentiles_plot.png
Saved DataFrame to ESP_init_forVEI_[2, 3, 4, 5, 6]/Forecast_MASS_FRACTION_ASH_BELOW_63_micron\forecast_MASS_FRACTION_ASH_BELOW_
63_micron.csv
       MER_kg/s  MASS_FRACTION_ASH_BELOW_63_micron_Percentile_5  \
0  5.370382e+05                                          0.136367
1  2.834315e+06                                          0.326842
2  1.739480e+07                                          0.845763
3  1.137084e+08                                          2.260471
4  7.590395e+08                                          6.105618

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_10  \
0                                         0.136545
1                                         0.327399
2                                         0.848057
3                                         2.268786
4                                         6.135787

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_15  \
0                                         0.136672
1                                         0.327787
2                                         0.849558
3                                         2.274889
4                                         6.157705

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_20  \
0                                         0.136770
1                                         0.328084
2                                         0.850797
3                                         2.279506
4                                         6.174201

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_25  \
0                                         0.136858
1                                         0.328337
2                                         0.851797
3                                         2.283505
4                                         6.188333

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_30  \
0                                         0.136933
1                                         0.328563
2                                         0.852696
3                                         2.287105
4                                         6.201066

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_35  \
0                                         0.137002
1                                         0.328753
2                                         0.853542
3                                         2.290274
4                                         6.212714

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_40  \
0                                         0.137068
1                                         0.328950
2                                         0.854379
3                                         2.293298
4                                         6.223760

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_45  \
0                                         0.137137
1                                         0.329149
2                                         0.855114
3                                         2.296171
4                                         6.234166

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_50  \
0                                         0.137200
1                                         0.329343
2                                         0.855869
3                                         2.299154
4                                         6.245058

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_55  \
0                                         0.137266
1                                         0.329530
2                                         0.856603
3                                         2.302091
4                                         6.255673

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_60  \
0                                         0.137332
```

```
1                                        0.329720
2                                        0.857362
3                                        2.305306
4                                        6.266952

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_65  \
0                                        0.137397
1                                        0.329911
2                                        0.858185
3                                        2.308274
4                                        6.278033

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_70  \
0                                        0.137465
1                                        0.330128
2                                        0.859049
3                                        2.311760
4                                        6.290510

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_75  \
0                                        0.137544
1                                        0.330335
2                                        0.859940
3                                        2.315143
4                                        6.303648

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_80  \
0                                        0.137626
1                                        0.330575
2                                        0.860961
3                                        2.319165
4                                        6.318528

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_85  \
0                                        0.137729
1                                        0.330837
2                                        0.862091
3                                        2.323896
4                                        6.334557

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_90  \
0                                        0.137844
1                                        0.331207
2                                        0.863684
3                                        2.330071
4                                        6.357026

   MASS_FRACTION_ASH_BELOW_63_micron_Percentile_95
0                                        0.138036
1                                        0.331769
2                                        0.865862
3                                        2.338660
4                                        6.389132
Results saved in folder: ESP_init_forVEI_[2, 3, 4, 5, 6]
Waiting for next VONA report...:   8%|          27:41
```